

Recursive Recovery of Negative Quantum Results: An Auditable Successor Protocol for Research Programmes

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Abstract

Scientific-agent systems increasingly preserve code, evidence chains and execution traces, but provenance alone does not determine what a research programme should do after a prospectively frozen hypothesis is refuted, absorbed by prior work or left `CANNOT_CHECK`. We present a bounded longitudinal case study of an executable **negative-result successor discipline** used in the ORION-Q quantum-compilation programme. Each research node retains its frozen claim, result receipt and disposition. A successor edge is admitted only when the predecessor outcome localizes a scientifically relevant responsibility, the strongest located donor receives first right of refusal, and a new protocol is frozen before its outcome. Later success never rewrites the predecessor. For publication, we define a 51-receipt frozen denominator: the 40 receipts in the original ORION-02 receipt index plus 11 pre-cut receipts needed to complete the later R6O–R6S/QG lineage. Twenty-three receipts are included as graph nodes and 28 are explicitly excluded with category-level reasons. The graph contains 13 asserted successor edges; seven negative/absorbed nodes are deliberately retained without an invented successor. A fail-closed validator checks the 51-receipt partition, binds every receipt to the publication cut, rejects graph drift or invented edges, and preserves negative-result visibility. Representative chains show restricted-family refutation to targeted family repair, second refutation to stronger support family, finite positive to all- n theorem, bounded prospective confirmation followed by later closed-form refutation, and candidate residual to donor absorption. We subtract direct prior art: ScienceAgentBench [1], AstaBench [2] and SciAgentArena [3] own substantial territory in rigorous scientific-agent evaluation; ScientistOne [4] owns broad claim-to-evidence traceability and integrity auditing. Our narrower object is the **authorized transition from a retained scientific disposition**

to the next frozen question. The evidence demonstrates feasibility and auditability in one exact-heavy programme; it does not establish improved scientific productivity, autonomous governance or cross-domain superiority.

0.1. Introduction

Scientific agents can search literature, write code, execute analyses and draft papers. Evaluation and verification have matured accordingly. **ScienceAgentBench** evaluates executable data-driven scientific tasks sourced from peer-reviewed work. **AstaBench** and **SciAgentArena** broaden controlled scientific-agent evaluation with standardized tools, costs, environments and task families. **ScientistOne** develops a Chain-of-Evidence framework in which claims, scores, references and method descriptions are explicitly audited against source evidence and implementation. These are direct donors. ORION-02 does not claim novelty for scientific-agent benchmarking, provenance, reproducibility, evidence chains or executable outcome scoring.

A different question begins after those controls return an unfavorable scientific outcome.

Suppose a hypothesis was frozen before execution and is then refuted. The programme may discard the run, reinterpret the claim, repeat the same mechanism with a new name, or preserve the negative but fail to use it. None of those choices says which new question is scientifically licensed by the observed failure.

ORION-02 studies a bounded alternative:

`frozen question -> result -> typed disposition -> responsibility/donor analysis -> frozen successor or stop.`

The successor is a new research object, not a repair of the predecessor. A later theorem can strengthen authority for a new claim without turning a finite predecessor into a theorem. A later counterexample can refute a broad extrapolation without invalidating an earlier bounded prospective success. A donor can absorb a candidate mechanism and terminate that novelty route without making the experiment scientifically useless.

ORION-Q is a useful case because many claims have exact referees or sharply frozen finite domains. That makes invalid succession easier to detect than in noisy empirical science. We therefore treat the programme as a **mechanism case study**, not as evidence that the discipline improves research productivity generally.

0.2. Donor boundary

0.2.1. Scientific-agent benchmarks

ScienceAgentBench (ICLR 2025) evaluates 102 real data-driven research tasks from 44 peer-reviewed publications with expert validation and executable program-level scoring. AstaBench and SciAgentArena provide additional controlled scientific-research benchmarks and environments. ORION-02 gives these lines full credit for rigorous task construction, tool/cost control, executable evaluation and benchmark methodology.

0.2.2. Evidence chains

ScientistOne directly owns broad claim-to-evidence traceability, score/reference verification and method-code consistency checking. ORION-02 assumes scientific states can be bound to evidence; its candidate residual starts only **after** that evidence has produced a disposition.

0.2.3. Preregistration/provenance/negative results

Prospective freezing, provenance and publication of negative results are established scientific practices. ORION-02’s candidate residual is their composition with two explicit transition constraints: responsibility-conditioned successor selection and donor first right of refusal. We make no “first” claim.

0.3. Research-state and successor model

A research node contains:

- claim/hypothesis and exact scope;
- protocol/outcome space fixed before result access;
- immutable result artifact;
- typed disposition such as SUPPORTED, NEGATIVE, REFUTED, DONOR_ABSORBED, CANNOT_CHECK, MIXED or OPEN;
- supported responsibility interpretation;
- donor/prior-work disposition;
- optional successor pointer.

A receipt proves identity/replay properties, not scientific truth or novelty. An asserted successor edge requires:

1. predecessor disposition exists;
2. the predecessor identifies a residual/failure layer relevant to the successor;

3. donor/prior alternatives are considered before candidate novelty is granted;
4. successor protocol is frozen after predecessor result and before successor outcome;
5. predecessor artifact/claim boundary remains unchanged;
6. successor may fail, be donor-absorbed or remain `CANNOT_CHECK` without retroactively changing the old result.

Chronological adjacency is not sufficient to create an edge.

0.4. Frozen denominator and machine-readable graph

A case-study narrative can cherry-pick clean recovery chains unless the selection universe is declared. ORION-02 therefore freezes a publication denominator in `Q2_ELIGIBLE_RECEIPT_INVENTORY_V1.json`.

0.4.1. Declared 51-receipt universe

The universe contains:

- 40 receipts from the original `RECEIPT_INDEX.md` at the publication cut;
- 11 additional **pre-cut** receipts needed to complete the later R6O–R6S and QG5/QG7 lineage plus the N2-F5B donor-first-refusal result.

The inventory partitions these 51 artifacts exactly into:

- **23 included graph nodes**;
- **28 explicit exclusions**, grouped as pre-ladder/orthogonal programme results, R6 instrument/control states, or N-lane results not needed for the selected donor-first-refusal examples.

Exclusion means “not a ORION-02 publication transition node,” not “scientifically unimportant.” The inventory does not claim that all 51 receipts should have a successor.

0.4.2. Graph denominator

`Q2_TRANSITION_GRAPH_V2.json` contains:

- 23 result nodes;
- 13 asserted successor edges;
- the main R6M/TARE recovery chain;

- donor-absorption/robustness examples;
- seven named negative/absorbed nodes deliberately retained without an asserted successor edge.

The paper therefore reports a **declared bounded graph**, not an unbounded history of every possible research transition.

0.4.3. *Fail-closed validator*

`check_transition_graph.py` runs in publication CI and checks:

- the 51-receipt universe is exactly partitioned 23/28;
- every denominator receipt exists at the publication cut and remains byte-identical;
- the graph's 23 artifact paths equal the included set exactly;
- node/edge IDs are unique and endpoints exist;
- required central-chain edges remain present;
- an edge carries both relation and reason;
- standalone negatives cannot silently acquire successor edges;
- negative/partial-result visibility cannot regress.

The validator does **not** establish scientific causality. It establishes that the paper's declared causal graph and denominator are reproducible, frozen and non-cherry-picked relative to the stated publication universe.

0.5. *Recovery chain I: common-anchor closure to split counterexample*

A restricted shared-Tag TARE family appears plausible on early exact/chemistry checks. A support-dominance analysis explicitly retains one unresolved coupling: changing a frame can change the Tag requirement.

A frozen hostile instance returns an exact counterexample: split anchors with a spread Tag cost 8 while the common-anchor family costs 9.

The result remains:

REFUTED: common-anchor closure.

The exact witness localizes the residual to Tag/anchor coupling. Only after that result does the programme freeze D+, admitting arbitrary anchors with exact compatible Tag minimization.

This is the simplest successor pattern:

exact refutation -> localized missing family coordinate -> prospectively frozen repair.

0.6. Recovery chain II: the repair is also refuted

D+ is attacked on a registered structured domain and fails by exact frame-for-Tag borrowing. The minimal witness has cost 5 while D+ costs 6.

The failure is not relabeled as another split-anchor case. Its witness identifies a different residual: support two can be profitable on a cheap central branch because it buys a cheaper Tag. That observation licenses a D++ support-two family.

Again the original D+ refutation remains visible.

0.7. Finite positive to all-**n** theorem

D++ matches the unrestricted optimum on all then-registered finite domains. The programme records this as a bounded positive, **not** a universal theorem.

The unresolved responsibility becomes proof-level: can support three or larger ever be necessary outside the checked domains? A separate theorem protocol attacks that question. R6S later proves all-**n** support-two sufficiency for the frozen grammar/objective.

The evidence hierarchy remains explicit:

- R6P: finite family closure;
- R6S: later all-**n** theorem with stronger authority.

The theorem does not retroactively change what the finite experiment established at the time.

0.8. Prospective confirmation followed by later refutation

A compact donor/split/borrow predictor is derived on frozen panels. A previously unread public Benzene subject is selected by a precommitted rule; its predictions are digest-stamped before exact truth is opened. All 15 matchings agree.

That result remains a valid **bounded prospective confirmation**.

Later QG5 produces a different exact **n=3** row where the simple closed form predicts 11 and exact truth is 10. The correct update is:

- earlier subject-level prediction remains correct in its scope;
- universal extrapolation is refuted;
- exact witness localizes a missing borrow shape;
- separately frozen B' tests that localization.

Later QG7 refutes B' with another support-two configuration; B'' is then frozen and closes its registered finite panels while the all-n smallest-family classification remains partially open.

A positive forecast and a later negative therefore coexist without contradiction because their scopes/authority differ.

0.9. Donor absorption as a successful research terminal

Other ORION-Q lanes show a distinct succession pattern. A candidate can be matched by a stronger donor/model-selection baseline. The correct disposition is `DONOR_ABSORBED`, not a renamed candidate win.

The graph deliberately includes donor-absorbed/negative nodes even when no successor edge is asserted. Their role is to constrain future work: repeating the same mechanism under new branding is not an admissible novelty route.

0.10. What ORION-02 demonstrates

The case study establishes three bounded properties.

Executable representation. Dispositions and successor edges can be represented in a machine-readable graph and validated against immutable artifacts.

Authority separation. Exact refutation, bounded positive, theorem, donor absorption and open state remain distinct rather than collapsing into one progress score.

Negative persistence. A later repair cannot make a predecessor disappear from the declared denominator/graph.

ORION-02 does **not** demonstrate faster discovery, better research quality, autonomous governance, general responsibility diagnosis in noisy empirical science, or universal novelty of the transition protocol.

Those would require new controlled/cross-domain evidence and are intentionally outside the current claim.

0.11. Reproducibility and availability

The publication bundle should include:

- `Q2_ELIGIBLE_RECEIPT_INVENTORY_V1.json`;
- `Q2_TRANSITION_GRAPH_V2.json`;
- canonical validator stdout;
- reviewer-facing table mapping each main-text edge to predecessor/successor protocols/receipts;

- publication-cut git identity;
- permanent archive identifier once actually deposited;
- explicit reuse licence before any “open source” wording is used.

The repository is publicly inspectable, but public visibility is not a reuse licence.

0.12. Limitations

One programme. Feasibility/auditability is demonstrated in ORION-Q, not cross-domain effectiveness.

Exact-heavy setting. Exact referees make responsibility easier to localize than in noisy sciences.

Declared publication universe. The 51-receipt denominator is a bounded ORION-Q publication selection universe, not every scientific event in ORION or a sample of research generally.

Selection rule is methodological. Inclusion/exclusion reasons are explicit, but ORION-02 does not claim every excluded result could never inform a different successor analysis.

No productivity outcome. There is no controlled comparison of discovery time, paper quality or scientific yield against another governance policy.

Human/AI composition. Humans and multiple AI systems may participate; ORION-02 is not an autonomous-agent superiority result.

External novelty remains empirical. Internal donor searches/receipts cannot self-certify publication novelty.

0.13. Discussion

The successor discipline changes the meaning of “progress.” An exact counterexample can open the right family extension. A donor absorption can close an unproductive novelty route. A finite positive can justify a theorem attempt without being restated as a theorem. A prospectively confirmed case can remain valid after a different instance refutes the broad extrapolation.

This differs from evidence-chain work in locus, not opposition. Evidence chains answer **what supports this claim?** ORION-02 asks **given the supported disposition, what research transition is now authorized?** Scientific-agent benchmarks ask whether systems solve tasks under controlled conditions. ORION-02 instead tracks how the research question itself changes after evidence arrives.

The 51-receipt denominator and fail-closed graph validator make that case study inspectable rather than anecdotal. They still do not prove that the policy is optimal. That stronger question—whether one successor discipline improves science relative to another—is successor research rather than a hidden assumption of this paper.

0.14. Conclusion

Negative results can be represented as more than failed outputs. In the ORION-Q case study, a frozen result becomes a typed persistent state that can authorize, block or redirect later research while preserving its original claim boundary. A 51-receipt declared universe is partitioned into 23 publication graph nodes and 28 explicit exclusions; the graph carries 13 asserted successor edges and retains negative/donor nodes without invented links.

The candidate contribution is deliberately narrower than provenance or scientific-agent benchmarking: **an auditable successor relation from retained scientific disposition to a newly frozen question, with donor first right of refusal and append-only negative history.**

Data and Code Availability

The declared Q2 publication universe contains 51 version-controlled result receipts, partitioned into 23 publication graph nodes and 28 explicit exclusions. A fail-closed validator binds the denominator and graph to the frozen publication cut. The source repository is publicly inspectable. A permanent archive identifier and explicit reuse licence will be inserted only after the corresponding release is deposited and authorized.

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